

Pythagorean Proof Gallery Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	pythagorean-proof-gallery

Big idea: Mathematical proof is not just a result; it is a standard for why a result must be true.

One-Minute Teacher Brief

Students compare visual, algebraic, and verbal proofs of the Pythagorean theorem and decide what makes each convincing.

Student Prompt

Inspect Pythagorean Proof Gallery Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Pythagorean Proof Gallery Stimulus A	Student-facing stimulus 1: Visual rearrangement proof of $a^2 + b^2 = c^2$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Pythagorean Proof Gallery Stimulus B	Student-facing stimulus 2: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Pythagorean Proof Gallery Reveal / Twist	Student-facing stimulus 3: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical proof is not just a result; it is a standard for why a result must be true.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Pythagorean Proof Gallery Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Pythagorean Proof Gallery Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	What makes mathematical proof different from empirical confirmation?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical proof is not just a result; it is a standard for why a result must be true.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: What makes mathematical proof different from empirical confirmation?

Setup Checklist

- Prepare three proof formats and do not identify one as best at the start.
- Ask students to rank proof cards using different criteria: certainty, explanation, elegance, and accessibility.

Example Stimuli or Materials

- Visual rearrangement proof of $a^2 + b^2 = c^2$.
- Algebraic proof card plus a verbal explanation card of the same theorem.

Resources To Use

- Resource pack: <resources/pythagorean-proof-gallery/index.html>
- Card deck CSV: <resources/pythagorean-proof-gallery/cards.csv>
- Visual prompt: <resources/pythagorean-proof-gallery/visual-prompt.svg>
- Proof evaluation grid: assumptions, conclusion, why it must be true, what it helps you understand.
- Criteria cards: rigorous, explanatory, elegant, memorable, generalisable.

Run Sequence

1. Display three different proofs of $a^2 + b^2 = c^2$.
2. Groups annotate what each proof shows, assumes, and leaves unclear.
3. Students rank proofs by convincingness, explanatory power, and elegance.
4. Groups compare whether their rankings changed across criteria.
5. Debrief why mathematical knowledge values proof rather than repeated measurement alone.

Debrief Moves

- Knowledge question: What makes a proof convincing?
- Can a visual proof explain better than an algebraic proof?
- Is elegance a mathematical virtue or an aesthetic judgement?

Teacher Quick Script

- A proof does more than persuade; it should show why the result must hold.
- Which proof gives certainty, and which proof gives understanding?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...